

Answer: C

15

$$d \sin \theta = n\lambda \Rightarrow d \sin(50^\circ) = 3\lambda \Rightarrow \frac{d}{\lambda} \approx 3.9162$$

For highest order,

$$\sin \theta < 1$$

$$\frac{n\lambda}{d} < 1 \Rightarrow n < 3.9162$$

highest order = 3

Answer: A

5

16 Work done needed is the total electrical potential energy at the centre of the square